Everett, I. H. Case, C. Murry, A. W. Crouch, F. C. Fosdick, Paul T. Payne and Otto Fluegel.

Detroit Lubricator Company, Detroit, Mich. Full line locomotive lubricators illustrating the oil control valve transfer system of filling lubricator No. 42, lubricator working under atmospheric hydrostatic pressure. Represented by A. D. Homard.

Dickerson Manufacturing & Supply Company, Clinton, Ill. Two improved safety water gages for steam boilers. Represented by C. L. Dickerson and R. R. Porterfield.

Elgin Watch Company, Chicago. Elgin railway watches. Represented by Cornelius J. Wiltsie.

Emery Pneumatic Lubricator Company, St. Louis, Mo. Brake cylinder lubricant; automatic lubricator for air brake equipment. Represented by N. J. McAloney.

Franklin Railway Supply Company, Franklin, Pa. Franklin pneumatic firedoor, vertically and horizontally opening types, grate shaker, McLaughlin flexible conduit, Franklin driving box lubricator, Franklin flexible metallic roof. Represented by W. L. Allison, R. G. Coburn, W. H. Coyle and Joseph Sinkler.

Galena Signal Oil Company, Franklin, Pa. Apparatus to show behavior of valves and cylinder oils in saturated and superheated steam at various temperatures in actual practice. Represented by P. H. Stack, W. Brumble, J. Graham, C. B. Royal, W. W. Lemon, L. H. Palmer, J. Eubank, M. M. Meham, L. Gleason, Robert McVicker, Edward McVicker, J. S. Brown, J. Linthen, W. J. Vance, J. Ferguson, J. A. Roosevelt, E. W. Hayes, J. A. Haher, R. E. Webb, George Barnes, Dr. P. H. Conradson, B. P. Corey, William Walsh and William Holmes.

Garlock Packing Company, Palmyra, N. Y. Complete line of the company's products, showing throttle valve and air pump packing; ring, spiral and coil packings for stationary power engines and for general use; sheet packing; special gaskets. Represented by J. P. Landreth, M. E. Hamilton and Phil Arnold.

Green-Tweed Company, New York. Palmetto and Manhattan packings, favorite reversible ratchet wrench and throttle packing. Represented by F. E. Ransley.

Hammett, G. H., Troy, N. Y. Showing Trojan metallic packing, Trojan bell ringers, triple valve bushing. Represented by E. C. Sawyer.

Hobart-Allfree Company, Chicago. Photographs and literature showing the Allfree locomotive valve gear. Represented by E. H. Allfree and W. H. England.

Hunt Spiller Manufacturing Corporation, Boston, Mass. Gun iron for locomotive castings, cylinder packing and valve packing for saturated and superheated engines, rod bushings, cross-head shoes, eccentric and straps, air pump bushing and knuckle joint bushing. Represented by J. G. Platt and V. W. Ellet.

Interlocking Nut and Bolt Company, Pittsburgh, Pa. Samples of bolts for track, locomotives, cars, automobiles and block signals. Represented by R. A. Clark.

Jacobs-Shupert Firebox Company, New York. Working model one-tenth size of Jacobs-Shupert sectional firebox, also ruptured segment removed from Santa Fe engine, type 3000, which went through case of low water, proving firebox could not be blown up; cross section full size of firebox. Represented by A. W. Whiteford.

Jenkins Bros., New York. J. B. Globe angle and check valves, special features of standard and extra heavy valves, Overland sheet packing, medium and extra heavy gate valves. Represented by B. J. Neeley.

Johns-Mansville Company, H. W., New York. Samples of packing and J.-M. air brake cylinder packing expanding ring. Represented by J. C. Younglove.

Leslie Company, The, Lyndhurst, N. J. Leslie steam heat regulator, Leslie electric headlight pressure regulator. Represented by S. Inglis Leslie and J. J. Cizek.

Liberty Manufacturing Company, Pittsburgh, Pa. Twin strainer for suction water lines, tube cleaners for stationary boilers and locomotive arch tubes, combined tube cutters and expanders. Represented by W. M. Burns.

Locomotive Improvement Company, Clinton, Iowa. Markels removable driving box brass, lateral motion plates, flangeless shoes and wedges, solid head back end main rod. Represented by L. W. Barker.

Locomotive Superheater Company, New York. Type A top header superheater. Represented by Geo. L. Bourne, W. G. Lawse, W. A. Buckbee, R. Osterman, H. B. Oatley and S. Hoffman.

Metford & Company, Chicago. Three-feed force-feed lubricator in operation, one-feed pumping against more than 1,000 lbs. pressure, one-feed lubricating a standard driving box, showing process of actually lifting box off journal and making it run in oil. Represented by R. L. McIntosh and W. J. Schlacks.

Manning, Maxwell & Moore, New York. Injectors, pop valves, steam gages, Hancock oil cups, boiler check valves, automatic cylinder cocks, prismatic water gages. Represented by Joseph H. Bush and C. L. Brown.

Marshall & Huschart Machinery Company, Chicago. Photographs of railway machine shop tools; descriptive literature and catalogs. Represented by N. W. Jones.

Mason Regulator Company, Boston, Mass. Regulating valves. Represented by Frank M. Morrison.

Midwestern Car Supply Company, Chicago. Anderson draft gear, full size and models. Represented by D. D. Ward.

Moon Manufacturing Company, Chicago. Electric headlight in operation; Moon turbine generator; Peerless arc headlight. Represented by H. E. Procunier and W. L. Procunier.

Nathan Manufacturing Company, New York. Injectors, lubricators, pop valves, water meter, Kellogg water alarm, clinger reflex water gages, boiler checks. Represented by C. A. Nathan, J. S. Seeley and George Royal.

National Boiler Washer Company, Chicago. Photographs of the various plants installed in the United States and Canada, and a description of the national low duty vacuum steam heating system. Represented by E. P. White, B. A. Brown and W. White.

New York Air Brake Company, New York. Sections of locomotive tender brakes, K triple valves, L triple valves. Represented by B. Pratt, L. W. Sawyer, Geo. Kleuger, F. Wentworth and B. Parker.

Ohio Injector Company, Chicago. Samples of Chicago lubricator and Ohio lifting injectors, Chicago non-lifting injectors, Chicago flange oilers. Represented by F. W. Furry, F. W. Edwards, W. S. Furry and A. C. Beckwith.

O'Malley-Beare Valve Company, Chicago. O'Malley-Beare multi-plate valves, gage cocks, Dudley blow-out valves. Represented by E. O'Malley, Thomas O'Malley and H. A. Crew.

Parkesburg Iron Company, Parkesburg, Pa. Charcoal iron boiler tubes; specimens of tubes showing various tests, photographs, etc. Represented by George Thomas, J. P. Mercer and J. H. Smythe.

Pyrene Engineering Company, Wilmington, Del. System of improved combustion for locomotives, eliminating smoke and saving coal. Represented by Wm. H. Sayer.

Paxton-Mitchell Company, Omaha, Neb. Metallic packing for piston rods and valves, steam and air pumps. Represented by C. A. Coons.

Pilliod Brothers Company, Toledo, Ohio. Working model of the Pilliod valve gear. Represented by Charles J. Pilliod.

Pilliod Company, The, Swanton, Ohio. Working model showing the Baker valve gear. Represented by Burton W. Mudge, Herbert Green, F. S. Wilcoxson and K. J. Eklund.

Pyle National Electric Headlight Company, Chicago. Locomotive electric headlights. Represented by J. W. Johnson, J. E. Kilker and H. P. Bayley.

Railway Age Gazette, New York. Copies of *Railway Age Gazette*, the Locomotive Dictionary, Railway Shop Kinks. Represented by L. B. Sherman, Samuel O. Dunn, Roy V. Wright, John N. Reynolds, Kenneth G. Cloud and Stanley Deacon.

Railway and Locomotive Engineering, Chicago. Copies of paper. Represented by Angus Sinclair and J. A. Cassell.

Railway List Company, Chicago. Copies of the *Master Mechanic* and *Official List of Railway Officials*. Represented my William E. McGraw.

Ryan & Johnson, Chicago. Models of coal passers and automatic slack adjusting drawbar between engine and tender. Represented by Y. J. Merriman, Edward Ryan, L. W. Galloway and Oscar Johnson.

Sargent Company, The, Chicago. Safety water gages and lubricator protectors, Osborne valves and high pressure joints, also E. S. E. blow-off valves. Represented by George Sargent and F. G. Dunbar.

Storrs Mica Company, Owego, N. Y. Samples mica lantern globes in ruby and white, samples mica headlight chimneys. Represented by Charles P. Storrs.

Street, Clement F., Schenectady, N. Y. Literature describing Street locomotive stoker. Represented by Clement F. Street and M. M. Lower.

Strong, Carlisle & Hammond, Cleveland, Ohio. Samples of Randall's graphite sheet lubricator; car brasses fitted with same. Represented by B. E. Carpenter, W. P. Thacher and W. P. Owens.

U. S. Metallic Packing Company, Philadelphia, Pa. Models of King packing for locomotive piston rods, valve stems and air pumps, air pump packing, lubricating device, Golmar bell ringer, oil cups, Leach sanders, Represented by Morris B. Brewster and E. Curtiss.

Watson-Stillman Company, New York. Full-sized working sectional model of Chambers throttle valve. Represented by Frank H. Clark.

Westinghouse Air Brake Company, Pittsburgh, Pa. Working demonstration of the centrifugal dirt collector, which takes the place of the brake pipe air strainer; educational charts showing No. 6 E T locomotive and tender brake equipment; improved K air brake triple valve; improved L passenger air brake triple valve; 8½-in. Cross compound air compressor. Represented by L. M. Carlton, F. B. Farmer, C. J. Olmsted, E. Hardenstein, S. J. Kidder, F. B. Johnson, C. C. Farmer, Lawrence Wilcox, C. H. Beck, J. A. O'Malley, F. H. Parke and W. M. Sleet.

Winslow Company, Horace L., Chicago. Blow-off valves, gages, Clark blow-off system for locomotives; circulars. Represented by Horace L. Winslow and T. H. Price.

International Railway General Foremen's Association.

The executive committee of the International Railway General Foremen's Association has outlined the following subjects for discussion at next year's convention: "How Can Shop Foremen Best Promote Efficiency?" to be presented by William G. Reyer, general foreman, Nashville, Chattanooga & St. Louis, Nashville, Tenn. (this will be a continuation of the paper presented at the convention in 1910); "Shop Supervision and Local Conditions," to be presented by W. W. Scott, general foreman, Cincinnati, Hamilton & Dayton, Indianapolis, Ind.; "Shop Specialization, Work and Tools," by W. T. Gale, demonstrator, Chicago & Northwestern, Chicago; "Roundhouse Efficiency," by William Hall, shop foreman, Chicago & Northwestern, Escanaba, Mich. L. H. Bryan, Duluth & Iron Range, Two Harbors, Minn., is secretary of the association.

American Peat Society.

The fifth annual meeting of the American Peat Society will be held at the American House, Kalamazoo, Mich., September 21-23. Reports will be made by Canadian officials on the commercial results of government fuel plants and peat gas producers in Canada. Papers will be presented on the drainage of peat bogs, on peat as fuel, peat gas and by-products, peat powder as a power factor, and various other products. Julius Bordollo, Kingsbridge, N. Y., is secretary.

MEETINGS AND CONVENTIONS.

The following list gives names of secretaries, dates of next or regular meetings, and places of meeting.

Air Brake Association.—F. M. Nellis, 53 State St., Boston, Mass.

American Association of Demurrage Officers.—A. G. Thomason, Scranton, Pa.

American Association of General Passenger and Ticket Agents.—C. M. Burt, Boston, Mass.; next meeting, St. Paul, Minn., Sept. 19, 1911.

American Association of Freight Agents.—R. O. Wells, East St. Louis, Ill.

American Association of Railroad Superintendents.—O. G. Fetter, Carew building, Cincinnati, Ohio; 3d Friday of March and September.

American Electric Railway Association.—H. C. Donecker, 29 W. 39th St., New York; October 9-13, Atlantic City, N. J.

American Railway Association.—W. F. Allen, 75 Church St., New York; November 15, Chicago.

SETTING INSIDE ADMISSION PISTON VALVES WITH WALSCHAERT GEAR.*

A design of Walschaert valve gear used with inside admission piston valves is shown in Fig. 1. The names of the various parts are as follows: *A*, valve; *B*, valve stem; *C*, combining lever; *D*, crosshead link; *E*, radius rod; *F*, reverse shaft; *G*, lifting link; *H*, reach rod; *K*, reverse lever; *L*, reverse link; *M*, eccentric rod; and *N*, eccentric crank. The main pin is shown on forward dead center and the reverse lever is in its middle position, with the link block in the center of the link. A careful study of the diagram shows that with a valve having inside admission the valve rod is connected to the combining lever at a point *below* the latter's connection to the radius rod. If the block is in the *lower* half of the link when in forward gear, the eccentric crank *follows* the main pin. If the block is in the *upper* half of the link when in forward gear, the eccentric crank *leads* the main pin.

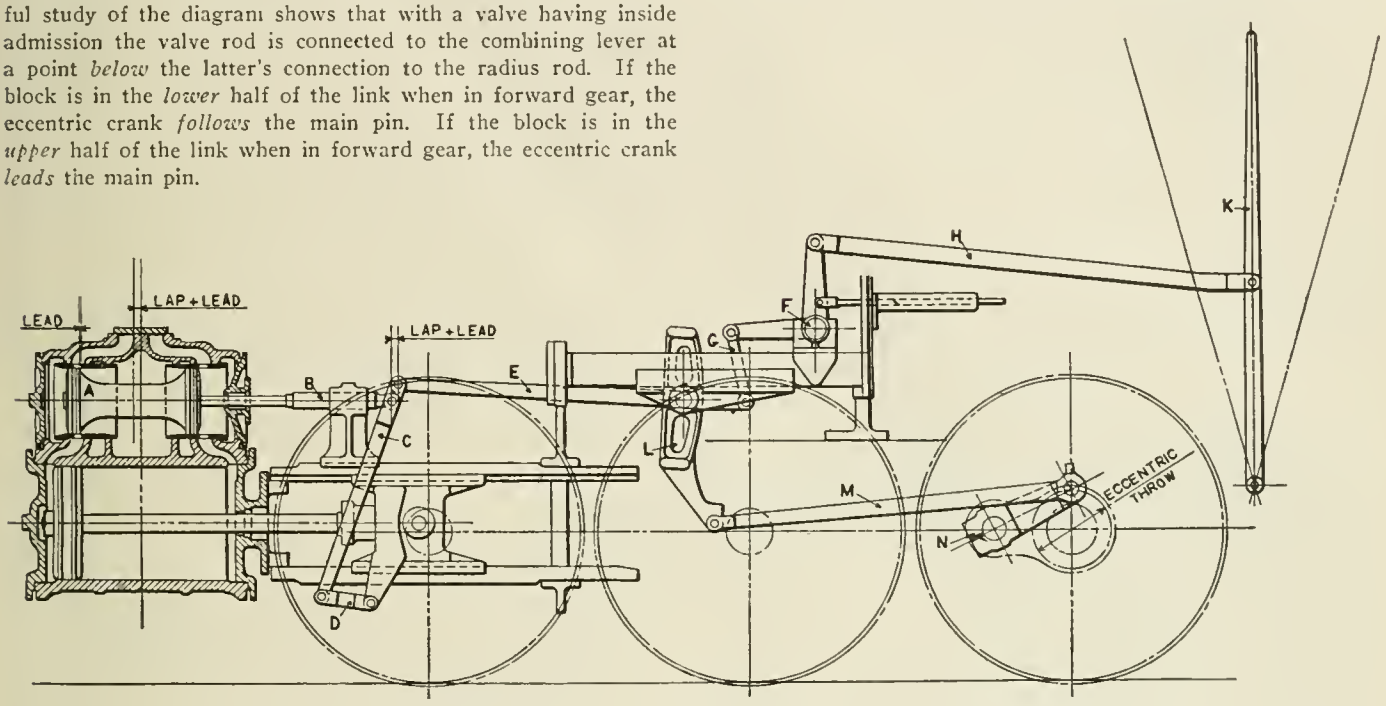


Fig. 1—Walschaert Valve Gear as Used with Inside Admission Piston Valves.

The method of setting inside admission piston valves is generally similar to that for outside admission valves, as described in the *Railway Age Gazette* of June 2, 1911, page 1254. It must be remembered, however, that to perform corresponding functions this valve moves in a direction opposite to that of the slide valve. When setting piston valves, the steam chest heads should be removed, for the sake of convenience. The line and line positions of the valve are determined by observation through peep holes provided for the purpose. In this way the points *F* and *F'* (Figs. 2 and 3) are located on the valve stem by tramming from any convenient point on the back wall of the steam chest. The lead points on the valve stem can thus be obtained by placing the crank on the dead centers, and again tramming from the steam chest wall (Fig. 4). The test for lead is made as described for the outside admission slide valves, the combining lever occupying positions as shown in Fig. 5. The lead in full gear, with this style of valve, is examined precisely as for outside admission valves.

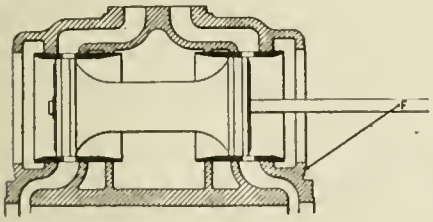


Fig. 2.

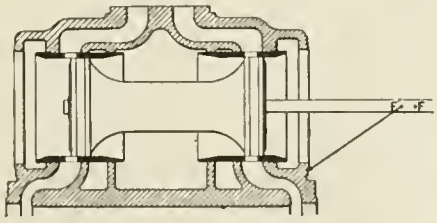


Fig. 3.

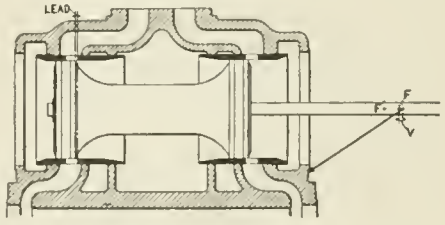


Fig. 4.

*From Record No. 70, published by the Baldwin Locomotive Works, Philadelphia, Pa., and entitled "Walschaerts Valve Gear."

CORRECTIONS.

Methods used for correcting errors can be best explained by two hypothetical cases. For example, suppose the specification of a locomotive having inside admission piston valves, calls for the following:

Maximum valve travel, 5¾ in.
Eccentric crank throw, 15½ in.
Constant lead, ¼ in.
Steam lap of valve, 1 in.
Link block below link center in forward gear.

The influence of the eccentric rod changes on the direction (ahead or back) of the movement of the valve, is explained by reference to Fig. 6. An examination of this figure will show that if the eccentric rod *E* is lengthened to *E'*, then the radius rod *R* will be moved ahead to the position *R'*, and the valve stem will be moved a distance *X* in the direction of the arrow, thus displacing the valve from position *V* to *V'*. Therefore, the rules applying in the case of outside admission slide valves also apply to this style of valve, as follows. In forward motion if the eccentric rod is lengthened, the valve is moved ahead. If the eccentric rod is shortened, the valve is moved back. In backward motion if the eccentric rod is lengthened, the valve is moved back. If the eccentric rod is shortened, the valve is moved ahead.

If the link block is above the center when running ahead, then, in each case, the valve will be moved in the direction opposite to that stated above.

In any case, regardless of whether the gear is in forward or backward motion, to move the valve ahead, lengthen the radius rod the amount desired. To move the valve back, shorten the radius rod the amount desired. It must be remembered that with an inside admission valve, the front port opening is in-